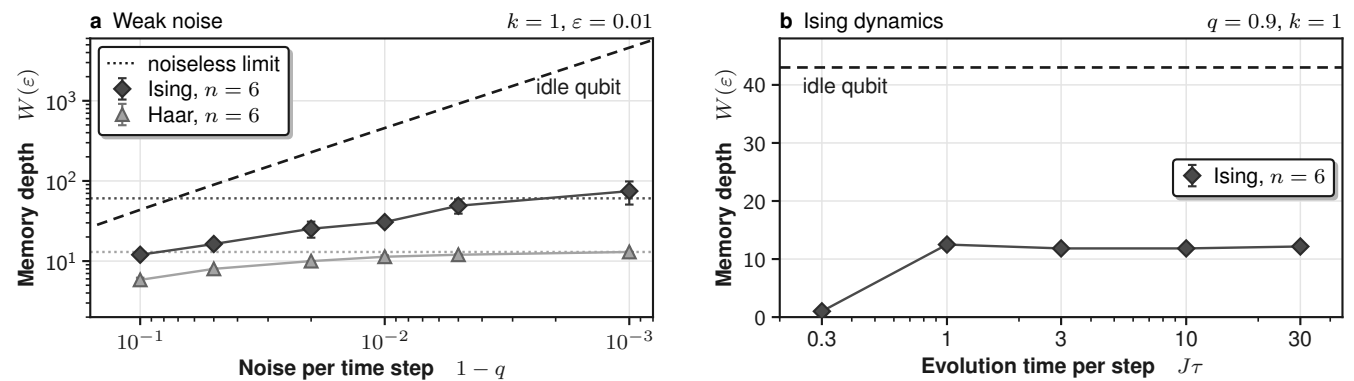

a Weak noise
$k = 1, \varepsilon = 0.01$
noiseless limit
Ising, $n = 6$
Haar, $n = 6$
idle qubit
Memory depth $W(\varepsilon)$
10^3
10^2
10^1
10^{-1}
10^{-2}
10^{-3}
Noise per time step $1 - q$
b Ising dynamics
$q = 0.9, k = 1$
idle qubit
Ising, $n = 6$
Memory depth $W(\varepsilon)$
40
30
20
10
0
0.3
1
3
10
30
Evolution time per step $J\tau$